

MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

DPP: 7

Time Speed Distance I

- Q1** Aman is traveling a total distance of 360 km, covering part of it at 48 kmph and the rest at 72 kmph. The time spent traveling at 72 kmph is 2 hours longer than the time spent at 48 kmph. Determine the total time Aman takes to complete the journey.
(A) 5 hours 15 mins
(B) 5 hours 36 mins
(C) 5 hours 54 mins
(D) 6 hours 10 mins
- Q2** Two distances are in the ratio of 2:3, and the time taken to cover these distances is in the ratio of 7:9. If the difference between the two speeds at which the distances are covered is 10 kmph, find the greater speed in kmph.
- Q3** In a 4x200-meter relay race, each of the 4 athletes runs 200 meters. If their respective average speeds are 15 m/s, 12 m/s, 10 m/s, and 8 m/s, determine the team's average speed for the entire 800-meter race.
(A) 10 m/s (B) 10.5 m/s
(C) 10.67 m/s (D) 11.33 m/s
- Q4** Two distances are traveled at speeds of 50 kmph and 75 kmph, respectively. The combined average speed over the entire journey is found to be 65 kmph. What is the ratio between the two distances?
(A) 2:3 (B) 3:2
(C) 4:9 (D) 9:4
- Q5** If two equal distances are covered at speeds x and y , let 'a' represent the average speed for total journey over these equal distances. Let 'b' represent the average speed when two equal time intervals are spent running at speeds x and y . Then, which of the following will necessarily be equal to 'ab'?
(A) $x + y$ (B) $x - y$
(C) xy (D) $x + y - xy$
- Q6** A cyclist starts from Town A at 6:00 am and reaches Town B at 12:00 pm the same day. Another cyclist starts from Town A at 7:00 am and reaches Town B at 11:00 am the same day. At what time do the two cyclists meet?
(A) 8:00 am (B) 8:30 am
(C) 9:00 am (D) 10:00 am
- Q7** A train stops for 12 minutes each hour during a 5-hour journey. The average speed of the train, including the halts, is recorded as 48 kmph. Determine the average speed of the train excluding the halts (in kmph).
- Q8** The speed of the Bugatti is 40 km/hr more than the speed of the Tesla. The time taken by Bugatti to cover 450 km is equal to the time taken by Tesla to cover 360 km. Find the speed of Bugatti.
(A) 200 km/hr (B) 220 km/hr
(C) 190 km/hr (D) 160 km/hr
- Q9** The return journey from Varanasi to Prayagraj was completed at a speed 200% higher than the speed during the onward journey from Prayagraj to Varanasi. Calculate by what percentage the average speed for the entire trip was lesser than the speed during the journey from Varanasi to Prayagraj.
(A) 33.33% (B) 50%
(C) 60% (D) 66.66%



- Q10** Meera and Aditi start running a 1-kilometer race at the same time. Meera initially runs at a constant speed of 6 m/s but later increases her speed to 10 m/s after covering a certain distance. Aditi, on the other hand, runs the entire race at a steady speed of 8 m/s. If they both cross the finish line simultaneously, determine the distance (in meters) Meera had run before increasing her speed.
- Q11** If a person travels for 3 hours at a speed 'a' kmph; and for 5 hours at a speed 'b' kmph, the average speed over the 8-hour journey is 40 kmph. However, if the person travels for 4 hours at 'a' kmph and 2 hours at 'b' kmph, the average speed becomes 47 kmph. Determine the average speed if the person travels for equal time intervals at speeds 'a' kmph and 'b' kmph.
 (A) 41 kmph (B) 42 kmph
 (C) 43 kmph (D) 43.5 kmph
- Q12** Adani's company office is 54 km away from his house. He usually goes to the office at a speed of V km/h. If he travels with 80% more speed, then it takes 18 minutes less to reach the office. What is the value of V ?
 (A) 60 (B) 80
 (C) 54 (D) 68
- Q13** A child leaves for her school at 7:30 am, walking at a speed of 5 km/h. At 7:50 am, her mother realizes the child has forgotten her admit card and after some time, starts cycling to catch up with her at a speed of 15 km/h. If the mother catches up with the child at 8:15 am, then at what time does the mother start cycling?
 (A) 7:52 am (B) 7:55 am
 (C) 8:00 am (D) 8:05 am
- Q14** From his laboratory, Prof. Hardy reaches home at 9:05 pm if he travels at 6 km/h. If he leaves his laboratory at the same time, he reaches home at 8:50 pm travelling at 15 km/h. If he leaves his laboratory at 8:20 pm, at what speed (in km/h) must he travel so that he reaches home exactly at 9:10 pm?
 (A) 3 km/h (B) 6 km/h
 (C) 9 km/h (D) 12 km/h
- Q15** A metro train departs Station X heading toward Station Y. Ten minutes later, another metro departs Station Y heading toward Station X. The first metro reaches Station Y 8 minutes after the second metro reaches Station X. If the speeds of the two metros are 60 km/h and 90 km/h respectively, find the distance between Station X and Station Y (in km).
- Q16** A bus blows horn twice at an interval of 24 seconds. A man is running towards the bus heard the sounds of the horns at an interval of 22 seconds. If the speed of sound is 330 m/s, what is the running speed of the man (in m/s)?
- Q17** A bus departs City A and after traveling 60 km, a tire puncture forces it to reduce its speed. As a result, it reaches its destination, City B, 30 minutes late. Had the puncture occurred 40 km further, the bus would have been late by only 20 minutes. Find the distance between City A and City B.
 (A) 120 km (B) 140 km
 (C) 160 km (D) 180 km
- Q18** Sheetal started her journey from Agra to Lucknow by a cab. The actual speed of the cab is 180 km/hr. If the driver of the cab reduces the speed of the cab by 30 km/hr, then she will arrive at Lucknow 15 minutes later than the scheduled time. Calculate the total distance from Agra to Lucknow.
 (A) 240 Km (B) 140 Km
 (C) 225 Km (D) 150 Km
- Q19** Kang is riding a One-wheeler from point X to Y, but he is returning by his Jeep. This way, he takes a total of 12 hours. If he goes and



returns by using his Jeep, then it takes 2 hours less to reach. Find the ratio of speeds of One-wheeler and Jeep.

- (A) 9:4 (B) 5:7
(C) 4:7 (D) 7:5

- Q20** A train covers 340 km in $3\frac{2}{5}$ hours with a uniform speed. The time taken, in hours, to cover 460 km with the same speed is:
(A) $4\frac{2}{5}$
(B) $3\frac{1}{5}$
(C) $4\frac{3}{5}$
(D) $3\frac{2}{5}$
- Q21** Alok starts from his school at speed of 40 km/h and takes 15 minutes more than the scheduled time to reach his coaching. If he travels at 36 km/h from his school then he takes 24 minutes more than the scheduled time to reach his coaching, then what is the distance between school and his coaching?
(A) 90 km (B) 18 km
(C) 36 km (D) 54 km
- Q22** Mr. Thompson and his assistant, start from their home and check the water level of a reservoir every day at 8 a.m. and then instantly return home together. One day, Mr. Thompson reached the reservoir an hour earlier i.e. at 7 a.m. and after checking the water level, started walking towards home. His assistant, who started from their home at the same time as usual, was on the way to the reservoir, met Mr. Thompson on the way, took a U-turn and they both returned home 30 minutes earlier than usual. How long (in minutes) did Mr. Thompson walk that day before meeting his assistant? [Provided that no time is lost in the checking process.]
- Q23** A cyclist departs from City A at 6 am and arrives in City B at 10:30 am. Beginning at 6:02 am, buses leave City B every 12 minutes,

heading towards City A. These buses cross paths with the cyclist as they move in the opposite direction. If the cyclist increases their speed to three times the original, and the departure interval between buses is also increased to three times the original, with all other conditions staying the same, how many buses will pass the cyclist along the route?

- Q24** The front wheel of a bicycle has a smaller radius than its rear wheel. The radii of the front and rear wheels are 18 cm and 24 cm, respectively. The bicycle covered a distance of 66 km in 1 hour and 40 minutes. Assuming the bicycle maintained a constant speed throughout the journey, find the difference in the number of revolutions per hour between the two wheels.
(A) 7350 (B) 7700
(C) 8400 (D) 8750
- Q25** An aircraft covers a journey of 500 km. However, if it covers a distance of d km at its regular speed and the remaining distance at $\frac{1}{4}$ th of its regular speed, it arrives 1 hour late. On the other hand, if it covers a distance of $(d + 60)$ km at $\frac{1}{4}$ th of its regular speed and the remaining distance at its regular speed, it is late by only 20 minutes. Find d and the aircraft's regular speed.
(A) 100, 1060 km/h (B) 90, 1140 km/h
(C) 80, 1260 km/h (D) 70, 1340 km/h
- Q26** Every day, I leave home at the same time and arrive at the office right on time. Today, I departed 8 minutes earlier than usual but drove 20% slower than my regular speed. Surprisingly, I reached the office 4 minutes later than the reporting time. Which of the following pairs can be the distance between my home and office and my driving speed today?
(A) 60 km, 75 km/h (B) 50 km, 60 km/h
(C) 40 km, 32 km/h (D) 32 km, 32 km/h



Q27 Three buses departed simultaneously from Varanasi to Lucknow along the same highway. The second bus traveled 30 km/h faster than the first bus and reached Lucknow two hours earlier than the first bus. The third bus arrived in Lucknow forty minutes after the second bus, covering half the time at the speed of the first bus and the other half at the speed of the second bus. What is the distance (in km) between Varanasi and Lucknow?

Q28 A woman walks at a speed of 6 kmph and reaches her workplace 11 minutes later than her scheduled time. When she increases her speed by 3 kmph, she arrives 6 minutes earlier than the scheduled time. A cyclist, who starts from the workplace at the same time the woman starts walking at 6 kmph, meets her on the way and gives her a ride, allowing her to reach the workplace exactly on time. Determine the cyclist's speed in kmph, assuming the time spent for the woman to get on the bicycle and for the cyclist to turn around is negligible.

(A) 9 kmph (B) 9.3 kmph

(C) 10.2 kmph

(D) 10.5 kmph

Q29 Two cyclists met at 1:00 PM at a park. One cycled north, and the other headed east. By 6:00 PM, they were 145 km apart. If the faster cyclist covered 5% more distance than the slower one, determine the speed (in km/h) of the faster cyclist.

Q30 Point P is positioned 130 km away from Q. Along the path from Q to P, there are two points, R and S, such that the points lie in the following order Q-R-S-P. The distance between Q and R is 30% of the total distance between Q and S, while the distance between P and S is 30% of the distance from R to P. Normally, Neha travels from Q to R at a speed of 12 kmph and from R to P at a speed of 25 kmph. However, today she traveled from Q to S at a speed equal to her usual average speed from Q to P and completed the remaining distance at a speed of 6 kmph. What was her average speed for the entire journey from Q to P today?

(A) 10 kmph

(B) 11 kmph

(C) 13 kmph

(D) 15 kmph



Answer Key

Q1 B
Q2 70
Q3 C
Q4 C
Q5 C
Q6 C
Q7 60
Q8 A
Q9 B
Q10 375
Q11 C
Q12 B
Q13 C
Q14 A
Q15 54

Q16 30
Q17 D
Q18 C
Q19 B
Q20 C
Q21 D
Q22 45
Q23 3
Q24 D
Q25 C
Q26 D
Q27 120
Q28 B
Q29 21
Q30 C



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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Let the time travelled at 48 kmph be t hours and let the time spent travelling at 72 kmph be $(t+2)$ hours. So

$$48(t) + 72(t + 2) = 360$$

$$\Rightarrow 120t + 144 = 360$$

$$\Rightarrow t = 1.8 \text{ hours}$$

So total time to complete the journey = $2t + 2 = 3.6 + 2 = 5.6$ hours = 5 hours 36 mins

Video Solution:



Q2 Text Solution:

Let the distances be $2x$ km and $3x$ km and let the time taken be $7y$ hours and $9y$ hours, so

$$\frac{3x}{9y} - \frac{2x}{7y} = 10 \left(\frac{3x}{9y} > \frac{2x}{7y}, \text{ so it will be the higher speed} \right)$$

$$\Rightarrow \frac{1}{21} \left(\frac{x}{y} \right) = 10$$

$$\Rightarrow \frac{x}{y} = 210 \text{ kmph}$$

$$\text{So the greater speed} = \frac{3}{9} \left(\frac{x}{y} \right) = 70 \text{ kmph}$$

Video Solution:



Q3 Text Solution:

Time taken by the four players will be

$$\frac{200}{15} \text{ s, } \frac{200}{12} \text{ s, } \frac{200}{10} \text{ s and } \frac{200}{8} \text{ s.}$$

$$\text{So average speed} = \frac{800}{\frac{200}{15} + \frac{200}{12} + \frac{200}{10} + \frac{200}{8}}$$

$$= \frac{800}{200 \left(\frac{8+10+12+15}{120} \right)}$$

$$= \frac{4 \times 120}{45}$$

$$= 10.67 \text{ m/s}$$

Video Solution:



Q4 Text Solution:

Let the distances be x and y , then

$$\frac{x}{50} + \frac{y}{75} = \frac{x+y}{65}$$

$$\Rightarrow \frac{3x+2y}{150} = \frac{x+y}{65}$$

$$\Rightarrow 39x + 26y = 30x + 30y$$

$$\Rightarrow 9x = 4y$$

$$\Rightarrow x : y = 4 : 9$$

Video Solution:



Q5 Text Solution:

In case of equal distances, average speed considering distance as d (a)=

$$\frac{\frac{2d}{x} + \frac{2d}{y}}{2} = \frac{2}{\frac{1}{x} + \frac{1}{y}} = \frac{2xy}{x+y}$$

In case of equal time intervals, considering

$$\text{time period as } t \text{ (b)} = \frac{t(x)+t(y)}{2t} = \frac{x+y}{2}$$

$$\text{Thus } ab = \frac{2xy}{x+y} \times \frac{x+y}{2} = xy$$

Video Solution:



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**Q6 Text Solution:**

Time taken by first cyclist travelling from town A to town B = 6 hours

Time taken by second cyclist travelling from town B to town A = 4 hours

Lets assume the distance between the two towns and make it a multiple of LCM of the two time periods to make it easy. So

Let the distance between the two towns to be 12x.

So speed of first cyclist = $2x$ /h

and speed of second cyclist = $3x$ /h

So distance travelled by first cyclist till 7 am = $2x$

So time required after 7 am = $2x/(3x - 2x) = 2$ hours

So the two cyclists will meet at 9:00 am.

Video Solution:**Q7 Text Solution:**

Since the train is travelling only for 48 minutes in an hour, so excluding stoppage time the train travels 48 km in 48 minutes, so

Average speed of train excluding the halts =

$$\frac{48}{48} \times 60 = 60 \text{ kmph}$$

(No use of the time period since the stoppage is constant across the time period)

Video Solution:**Q8 Text Solution:**

Let speed of Bugatti = $(X + 40)$ km/hr

The speed of Tesla = X km/hr

The time taken by Bugatti to cover 450 km is equal to the time taken by Tesla to cover 360 km. We can use the formula:

Time = Distance / Speed

Since the times are equal, we can set up the equation:

$$\frac{360}{X} = \frac{450}{X + 40}$$

$$\Rightarrow 5X = 4X + 160$$

$$\Rightarrow X = 160$$

Hence, the speed of the Bugatti = $160 + 40 = 200$ km/hr.

Alternatively,

As the time taken is the same, the ratio of speeds will be the ratio of distances covered. So the ratio of speeds of Bugatti and Tesla will be 450 : 360 i.e. 5 : 4

Given the difference between speeds is 40 kmph, i.e. $5 - 4 = 1$ unit = 40 kmph

Speed of Bugatti = 5 units = $40 \times 5 = 200$ kmph

Video Solution:**Q9 Text Solution:**

Average speed when the travelling distance is same while travelling at different speeds =

$$\frac{2xy}{x+y}$$

Lets consider the speed while travelling from Prayagraj to Varanasi to be x kmph, hence the



speed for the return journey = $3x$
So the average speed = $\frac{2 \times x \times 3x}{x+3x} = 1.5x$
So the percentage by which the average speed for the entire trip was lesser than the speed of the journey from Varanasi to Prayagraj =
 $\frac{3x-1.5x}{3x} \times 100 = 50\%$

Video Solution:



Q10 Text Solution:

Let the distance ran by Meera at 6 m/s be x meters, s
 $\frac{x}{6} + \frac{1000-x}{10} = \frac{1000}{8}$
 $\Rightarrow \frac{5x+3000-3x}{30} = 125$
 $\Rightarrow 2x = 750$
 $\Rightarrow x = 375\text{ m}$
Alternatively,
Since the average speed of Meera would be 8 m/s only, she would have covered both these distances for an equal period of time.
Time take Aditi takes = $\frac{1000}{8} = 125\text{ seconds}$
Thus, she would have travelled for 62.5 seconds at a speed of 6 m/s
So the distance covered will be
 62.5×6
 $= 375\text{ m}$

Video Solution:



Q11 Text Solution:

Given

$$\frac{3a+5b}{8} = 40$$
$$\Rightarrow 3a + 5b = 320$$

Also

$$\frac{4a+2b}{6} = 47$$
$$\Rightarrow 4a + 2b = 282$$

So adding the two

$$7a + 7b = 602$$
$$\Rightarrow \frac{a+b}{2} = 43\text{ kmph}$$

So the average speed when the person travels equal time intervals at speeds a and $b = 43$ kmph

Video Solution:



Q12 Text Solution:

Let the actual time taken = t hours
We know, distance covered = speed $\times$ time
Now, $54 = V \times t$ (i)
Also, $54 = (100 + 80)\%$ of $V \times (t - \frac{18}{60})$
 $54 = \frac{9V}{5} \times (t - 0.3)$ (ii)
From (i) and (ii), we get
 $Vt = \frac{9V}{5} \times (t - 0.3)$
 $t = 1.8t - 0.54$
 $0.8t = 0.54$
 $t = \frac{27}{40}$
And, $54 = V \times \frac{27}{40}$
 $V = 80\text{ km/h}$
Alternatively,
Speed increases by 80% i.e. becomes 180% or $\frac{9}{5}$ times the original speed, so to cover the same distance $\frac{5}{9}$ of the original time will be required
Thus, $\frac{4}{9}$ of the original time will be saved
Given that,
 $\frac{4}{9}$ of original time = 18 minutes



$$\begin{aligned}\text{Original time} &= \frac{18 \times 9}{4} = \frac{81}{2} \text{ minutes} = \\ &= \frac{81}{2 \times 60} = \frac{81}{120} \text{ hours} \\ \text{Thus, } V &= \frac{54}{\frac{81}{120}} = \frac{54 \times 120}{81} = 80 \text{ kmph}\end{aligned}$$

Video Solution:



Q13 Text Solution:

Distance travelled by the child till 8:15 am =

$$\frac{45}{60} \times 5 = \frac{15}{4} \text{ km}$$

Time taken by mother to catch the child =

$$\frac{\frac{15}{4}}{\frac{4}{15}} \times 60 = 15 \text{ min}$$

So the mother starts cycling at 8:00 am

Video Solution:



Q14 Text Solution:

The timings of reaching the home at the speeds of 6 km/h and 15 km/h, are 9:05 PM and 8:50 PM, respectively.

Hence, the difference in the time taken = 15 minutes

Let the distance from the home to the lab be x km.

So, we have the relation to calculate time as:

$$\frac{x}{6} - \frac{x}{15} = \frac{15}{60}$$

$$\Rightarrow \frac{x}{10} = \frac{1}{4}$$

$$\Rightarrow x = \frac{5}{2}$$

$$x = 2.5 \text{ km}$$

He needs to reach the home by travelling for (8:20 pm to 9:10 pm) = 50 minutes

$$\text{The required speed} = \frac{2.5}{\frac{50}{60}} = 3 \text{ km/h}$$

Alternatively,

$$\text{Total Distance} = \frac{15 \times 6}{15 - 6} \times \frac{15}{60} = \frac{5}{2} \text{ km}$$

This distance is required to be covered in a

total of 50 minutes i.e. $\frac{5}{6}$ hours

So, the required speed will be

$$\frac{\frac{5}{2}}{\frac{5}{6}} = \frac{5}{2} \times \frac{6}{5} = 3 \text{ kmph}$$

Video Solution:



Q15 Text Solution:

If time taken by the metro travelling from station Y towards station X is t minutes, then time taken by the other metro will be (t+18) minutes.

So

$$\frac{60}{90} = \frac{t}{t+18}$$

$$\Rightarrow 2t + 36 = 3t$$

$$\Rightarrow t = 36 \text{ minutes}$$

So distance between the two stations =

$$90 \times \frac{36}{60} = 54 \text{ km}$$

Video Solution:



Q16 Text Solution:

In 2 s, sound travels $330 \text{ m/s} \times 2 \text{ s} = 660 \text{ m}$.

The time compression from 24 s to 22 s would happen if the man is 660 m closer to the bus



when the sound of the 2nd horn reached it than it was when the 1st horn reached it. So, the man's speed was

$$\frac{660}{22} = 30 \text{ m/s.}$$

Video Solution:



Q17 Text Solution:

The extra time taken by the bus will be directly proportional to the distance travelled by the bus in the punctured state.

So

$$\begin{aligned} & \frac{\text{distance travelled in punctured state in first case}}{\text{distance travelled in punctured state in second case}} \\ &= \frac{30}{20} \\ \Rightarrow & \frac{\text{distance travelled in punctured state in first case}}{\text{distance travelled in punctured state in second case}} \\ &= \frac{3}{2} \end{aligned}$$

So if the distance travelled in punctured state in first case is $3x$ km then in the second case it is $2x$ km, so

$$3x - 2x = 40 \text{ km}$$

$$\Rightarrow x = 40 \text{ km}$$

So the distance between the two cities =
 $60 + 3(40) = 180 \text{ km}$ or $100 + 2(40) = 180 \text{ km}$

Video Solution:



Q18 Text Solution:

The actual speed of the cab = 180 Km/hr
 The new speed of the cab after reducing the speed 30 Km/hr
 $= (180 - 30) = 150 \text{ Km/hr}$
 Thus,

Initial : New

Speed $180 : 150$

Rate of Time $150 : 180$

Time Left $5a : 6a$

Speed and time are inversely proportional to each other when distance is constant. So, if the speeds are in ratio $m:n$ then time taken will be in ratio $n:m$.

Thus, the actual time = $5a$ and new time = $6a$

Now, according to the question

$$(6a - 5a) = 15 \text{ min}$$

$$a = 15 \text{ min}$$

So, new time taken by Sheetal = $6a$

$$= 6 \times 15 = 90 \text{ min}$$

$$= \frac{3}{2} \text{ Hour}$$

So, the distance covered by Sheetal:

$$D = S \times T$$

$$= 150 \times \frac{3}{2} = 75 \times 3 = 225 \text{ Km}$$

Video Solution:



Q19 Text Solution:

Let the speed of his One-wheeler = v_1

Speed of his Jeep = v_2

The distance between X and Y = a

According to question,

$$\frac{a}{v_1} + \frac{a}{v_2} = 12 \dots\dots\dots (1)$$

$$\text{and } \frac{2a}{v_2} = 10, \text{ therefore } \frac{a}{v_2} = 5$$

$$\text{and } \frac{a}{v_1} = 7 \text{ (putting } \frac{a}{v_2} \text{ in eq. (1))}$$

$$v_1 = \frac{a}{7} \text{ and } v_2 = \frac{a}{5}$$

Therefore, the required ratio

$$= \frac{a}{7} : \frac{a}{5}$$

$$= 5 : 7$$

Alternatively,

While going by one-wheeler and returning by Jeep, it took a total of 12 hours



But when he went and returned by Jeep, it took a total of 10 hours

Thus, the jeep can cover one way journey in

$$\frac{10}{2} = 5 \text{ hours}$$

So the one-wheeler can cover one way journey in $12 - 5 = 7$ hours

As the distance is equal, and the time taken by one-wheeler and jeep is in the ratio 7 : 5, so the speeds will be in the ratio 5 : 7

Video Solution:



Q20 Text Solution:

$$\text{Speed of train} = \frac{340}{3\frac{2}{5}} = \frac{340}{\frac{17}{5}} \text{ kmph} = 100 \text{ kmph}$$

Now, distance = 460 km

$$\text{Required time} = \frac{460}{100} = \frac{23}{5} = 4\frac{3}{5} \text{ hours.}$$

Alternatively,

As the speed is constant, the ratio of the distances travelled will be equal to the ratio of the times taken.

Let the time required to cover 460 km with the given speed be 'x' hours, hence,

$$\frac{340}{460} = \frac{\frac{17}{5}}{x}$$

$$\frac{17}{23} = \frac{17}{5x}$$

$$x = \frac{23}{5} = 4\frac{3}{5} \text{ hours}$$

Video Solution:



Q21 Text Solution:

Let distance between school and his coaching = d km and time taken = t hours

$$\text{So, } \frac{d}{40} = t + \frac{15}{60}$$

$$\Rightarrow t = \frac{d}{40} - \frac{1}{4}$$

$$\text{And } \frac{d}{36} = t + \frac{24}{60}$$

$$\Rightarrow \frac{d}{36} = \frac{d}{40} - \frac{1}{4} + \frac{2}{5}$$

$$\Rightarrow \frac{d}{360} = \frac{(8-5)}{20}$$

$$\Rightarrow d = 360 \times \frac{3}{20}$$

$$\Rightarrow d = 54$$

Alternatively,

Total distance =

$$\frac{40 \times 36}{40 - 36} \times \frac{24 - 15}{60}$$

$$= \frac{40 \times 36}{4} \times \frac{9}{60}$$

$$= 54 \text{ km}$$

Video Solution:



Q22 Text Solution:

We see that his assistant reduces his total travel time by 30 minutes.

So, his assistant reduces its **one-way**

travelling time by $\frac{30}{2} = 15$ minutes.

Therefore, the assistant would have met Mr. Thompson 15 minutes before 8:00 a.m. i.e. 7:45 a.m.

This means that Mr. Thompson walked from 7 a.m. to 7.45 a.m., i.e., 45 minutes.

Video Solution:



Q23 Text Solution:

If the speed triples, the cyclist will reach city B at 7:30 AM.



Now the buses will leave after every 36 minutes, so they will leave at 6:02, 6:38, 7:14, 7:50, and so on. So 3 buses will pass the cyclist along the route.

Video Solution:



Q24 Text Solution:

Difference in the number of revolutions per minute between the two wheels

$$= \frac{66 \times 1000 \times 100}{2 \times \frac{22}{7} \times 18} - \frac{66 \times 1000 \times 100}{2 \times \frac{22}{7} \times 24}$$

$$= \frac{5}{3}$$

$$= \frac{66 \times 1000 \times 100 \times 3 \times 7}{2 \times 22 \times 5} \left(\frac{1}{18} - \frac{1}{24} \right)$$

$$= 3 \times 100 \times 100 \times 3 \times 7 \times \left(\frac{1}{72} \right)$$

$$= 8750$$

Video Solution:



Q25 Text Solution:

Let the normal speed of the aircraft be v km/h, and let the distance traveled at the normal speed in the first case be d km.

Case 1:

The aircraft travels d km at normal speed v , so time taken = $\frac{d}{v}$ hours.

The remaining distance is $(500 - d)$ km, and it is traveled at $\frac{1}{4}$ of the normal speed, i.e., $\frac{v}{4}$

Time taken for the remaining part = $\frac{500-d}{v/4} = \frac{4(500-d)}{v}$ hours.

In total, the time taken in this scenario is, $\frac{d}{v} + \frac{4(500-d)}{v}$ and the aircraft is late by 1 hour. So the total time exceeds the normal time by 1 hour. Thus, we have the equation:

$$\frac{d}{v} + \frac{4(500-d)}{v} = \frac{500}{v} + 1$$

$$d + \frac{4(500-d)}{1} = 500 + v$$

$$3d + v = 1500 \dots (i)$$

Case 2:

The aircraft now travels $(d + 60)$ km at $\frac{1}{4}$ of its normal speed, so time taken = $\frac{4(d+60)}{v}$ hours.

The remaining distance is

$500 - (d + 60) = 440 - d$ km, and it is travelled at normal speed v , so time taken = $\frac{440-d}{v}$ hours.

The total time in this scenario is $\frac{4(d+60)}{v} + \frac{440-d}{v}$ and the aircraft is late by 20 minutes,

i.e., $\frac{1}{3}$ rd of an hour. So we get the equation:

$$\frac{4(d+60)}{v} + \frac{440-d}{v} = \frac{500}{v} + \frac{1}{3}$$

$$12(d + 60) + 3(440 - d) = 1500 + \frac{v}{3} \dots$$

$$9d - v = -540 \dots (ii)$$

By solving (i) and (ii), we get

$$v = 1260 \text{ km/h}$$

$$d = 80 \text{ km}$$

Video Solution:



Q26 Text Solution:

Let the usual speed be x km/hr and the distance between the office and home be y km.

Given, at 20% slower speed ($\frac{4}{5}x$), it takes 12 minutes ($\frac{1}{5}h$) more to cover the distance of y km, so

$$\frac{y}{x} + \frac{1}{5} = \frac{y}{\frac{4}{5}x}$$

$$\Rightarrow \frac{5y+x}{5x} = \frac{5y}{4x}$$

$$\Rightarrow 20y + 4x = 25y$$

$$\Rightarrow x = \frac{5}{4}y$$

So

$$\frac{4}{5}x = \frac{4}{5} \times \frac{5}{4}y = y$$

So the reduced speed and distance between the office and home is same.

Video Solution:

**Q27 Text Solution:**

Let the speed of the first bus be x kmph and time required for it to travel be t , so speed of second bus will become $(x+30)$ kmph and time taken by it will be $(t-2)$ hours.

So

$$xt = (x + 30)(t - 2)$$

$$\Rightarrow xt = xt + 30t - 2x - 60$$

$$\Rightarrow t = \frac{x+30}{15}$$

For the third bus,

$$xt = x\left(\frac{t}{2} - \frac{2}{3}\right) + (x + 30)\left(\frac{t}{2} - \frac{2}{3}\right)$$

$$\Rightarrow xt = (2x + 30)\left(\frac{3t-4}{6}\right)$$

$$\Rightarrow 6xt = 6xt - 8x + 90t - 120$$

$$\Rightarrow 90t = 8x + 120$$

$$\Rightarrow 6(x + 30) = 8x + 120$$

$$\Rightarrow 6x + 180 = 8x + 120$$

$$\Rightarrow x = 30 \text{ kmph}$$

So

$$t = \frac{30+30}{15} = 4 \text{ hours}$$

So distance = $xt = 120$ km

Video Solution:

**Q28 Text Solution:**

Let the distance required to be covered by the woman to reach the workplace be x km, so
So

$$\frac{x}{6} - \frac{x}{9} = \frac{17}{60}$$

$$\Rightarrow \frac{x}{18} = \frac{17}{60}$$

$$\Rightarrow x = 5.1 \text{ km}$$

Time required by her to reach

$$\text{office on time} = \left(\frac{5.1}{6} \times 60\right) - 11$$

$$= 40 \text{ minutes}$$

So the cyclist will meet the woman

after 20 minutes (half time of

travelling to and fro).

So if the speed of the cyclist is c km/h, then

$$\frac{5.1}{6+c} = \frac{20}{60}$$

$$\Rightarrow 15.3 = 6 + c$$

$$\Rightarrow c = 9.3 \text{ km/h}$$

Video Solution:



**Q29 Text Solution:**

Since one of them is headed North and the other one is headed East, they will form arms of a right angle and hence the distance between them is the hypotenuse for the same.

Since the faster cyclist covered 5% more distance, it means his speed is 5% faster than the other one.

Let the speed of the slower cyclist be x km/h, so the speed of the faster cyclist = $1.05x$ =

$$\frac{21}{20}x \text{ km/h}$$

So distance covered by them in 5 hours will be

$$5x \text{ and } \frac{105}{20}x$$

So

$$(5x)^2 + \left(\frac{105}{20}x\right)^2 = (145)^2$$

$$\Rightarrow x^2 + \left(\frac{21}{20}x\right)^2 = (29)^2$$

$$\Rightarrow \frac{400x^2 + 441x^2}{400} = 841$$

$$\Rightarrow 841x^2 = 841 \times 400$$

$$\Rightarrow x = \sqrt{400} = 20 \text{ km/h}$$

So

$$\text{Speed of the faster cyclist} = \frac{21}{20} \times 20$$

$$= 21 \text{ km/h}$$

Video Solution:

**Q30 Text Solution:**

Ratio of lengths

$$QR:QS = 3:10$$

So

$$QR : RS = 3:7$$

Also

$$PR : PS = 10:3$$

So

$$RS : PS = 7:3$$

So

$$QR:RS:PS = 3:7:3$$

So

$$QR = 30 \text{ km}$$

$$RS = 70 \text{ km}$$

$$PS = 30 \text{ km}$$

Average speed of Neha from Q to P =

$$\frac{\frac{130}{\frac{30}{12} + \frac{100}{25}}}{\frac{130}{6.5}} = 20 \text{ km/h}$$

So, average speed of Neha in the new scenario

$$= \frac{130}{\frac{100}{20} + \frac{30}{6}} = \frac{130}{10} = 13 \text{ km/h}$$

Video Solution:



[Android App](#)

| [iOS App](#)

| [PW Website](#)